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UNIVERSITY OF CAMBRIDGE INTERNATIONAL EXAMINATIONS
General Certificate of Education Ordinary Level

PHYSICS **5054/13**

Paper 1 Multiple Choice **October/November 2010**

1 hour

Additional Materials: Multiple Choice Answer Sheet
Soft clean eraser
Soft pencil (type B or HB is recommended)

0753457733

READ THESE INSTRUCTIONS FIRST

Write in soft pencil.
Do not use staples, paper clips, highlighters, glue or correction fluid.
Write your name, Centre number and candidate number on the Answer Sheet in the spaces provided unless this has been done for you.

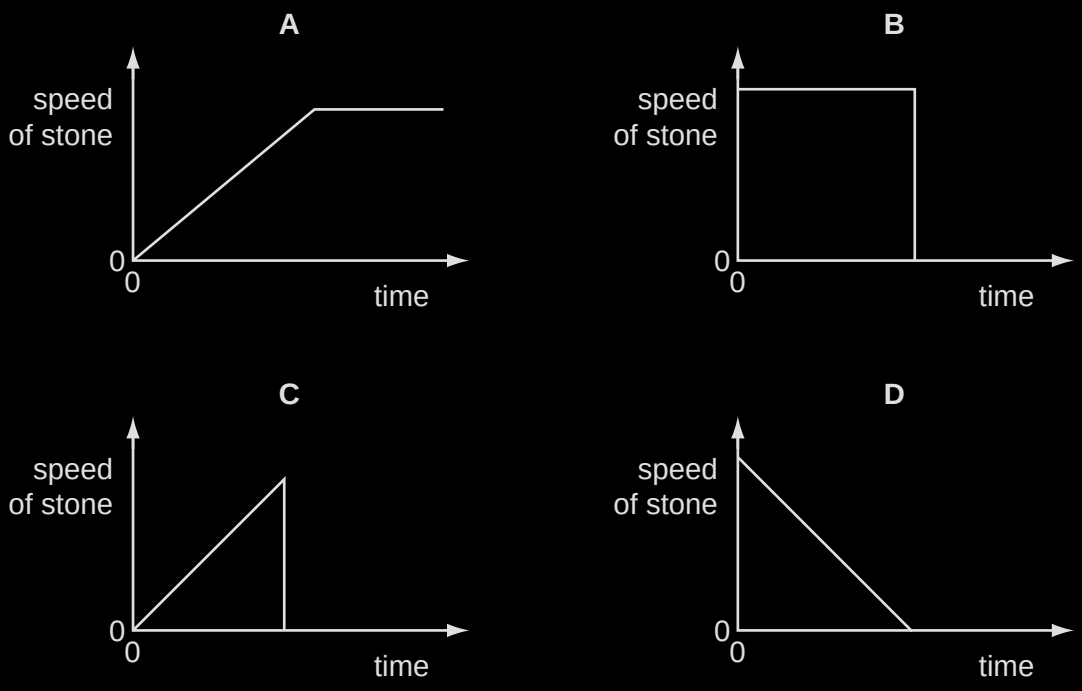
There are **forty** questions on this paper. Answer **all** questions. For each question there are four possible answers **A**, **B**, **C** and **D**.
Choose the **one** you consider correct and record your choice in **soft pencil** on the separate Answer Sheet.

Read the instructions on the Answer Sheet very carefully.

Each correct answer will score one mark. A mark will not be deducted for a wrong answer.
Any rough working should be done in this booklet.

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3 A car travels along a road. The driver stops the car by pushing his foot down on the brake pedal.

What does **not** change if he pushes harder on the brake pedal?

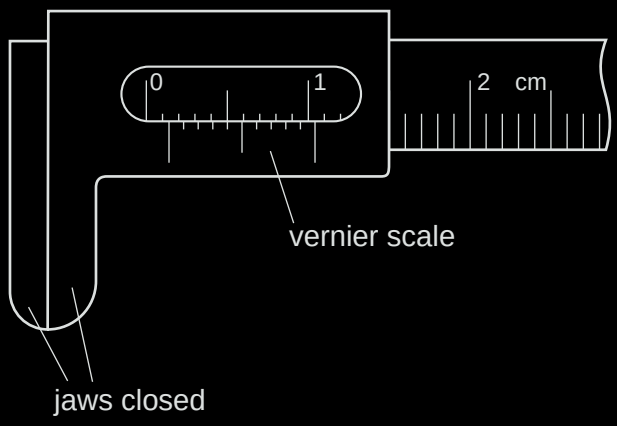
A the braking distance

B the braking force

C the stopping distance

D the thinking distance

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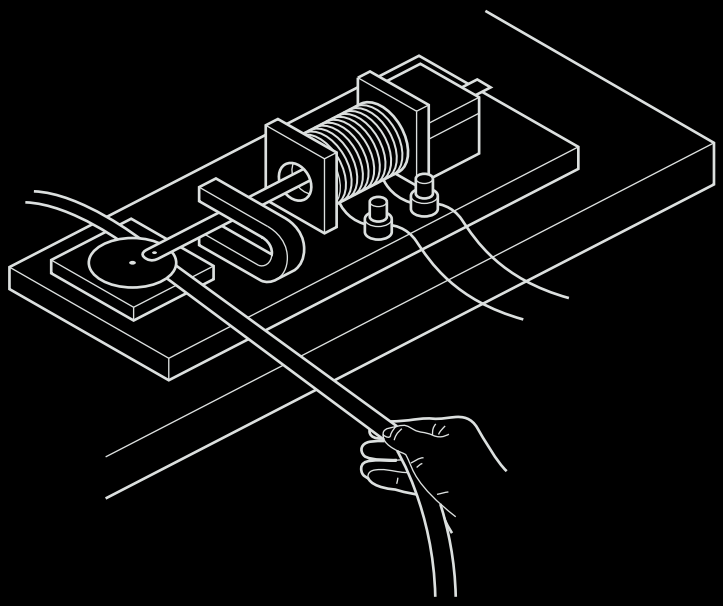


What is the zero error?

A 0.04 cm **B** 0.05 cm **C** 0.14 cm **D** 0.15 cm

5 A student pulls a piece of tape through a ticker-tape timer. Every 0.02 s, the timer prints a dot on the tape.

First the tape is pulled quickly, then slowly, then quickly again.



Which piece of tape does the student obtain?

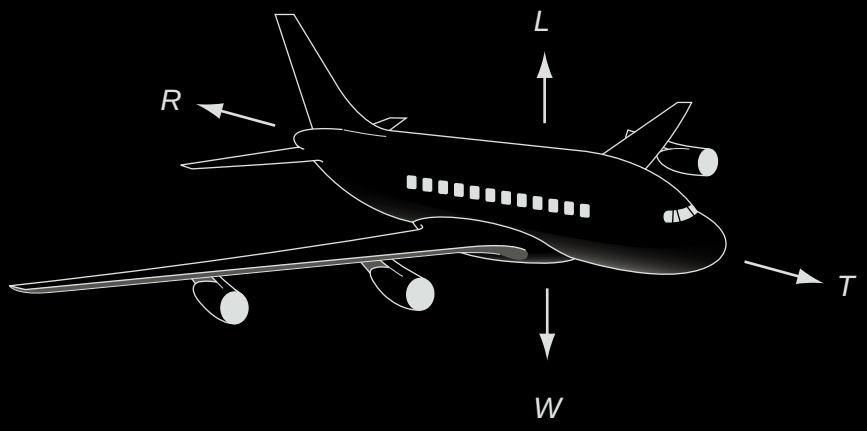
A

B

C

D

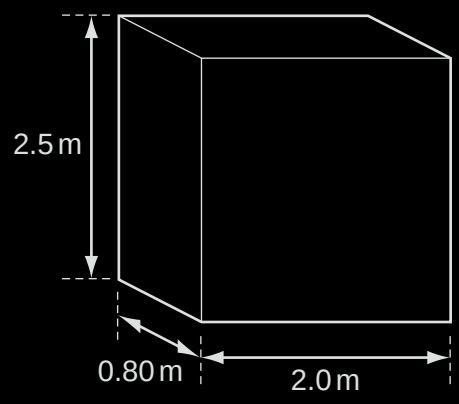
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Which row is correct?

	vertical forces	horizontal forces
A	$L = W$	$T = R$
B	$L > W$	$T > R$
C	$L = W$	$T > R$
D	$L > W$	$T = R$

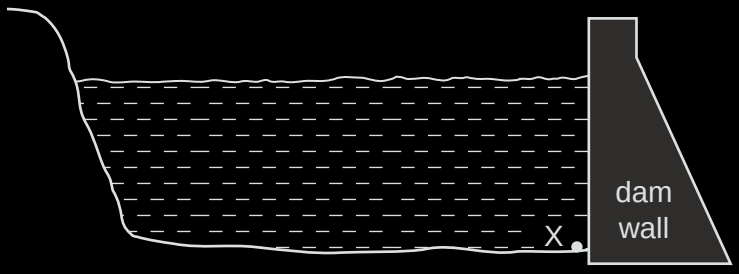
7 The base for a statue rests on level ground. It is made from stone and is 2.0 m long, 2.5 m high and 0.80 m wide. It has a weight of 96 000 N.



What is the pressure that the base exerts on the ground?

A 19 kPa **B** 24 kPa **C** 48 kPa **D** 60 kPa

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Which factor determines the pressure at X?

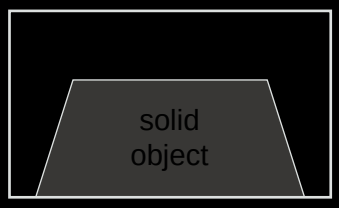
A the depth of the water in the reservoir

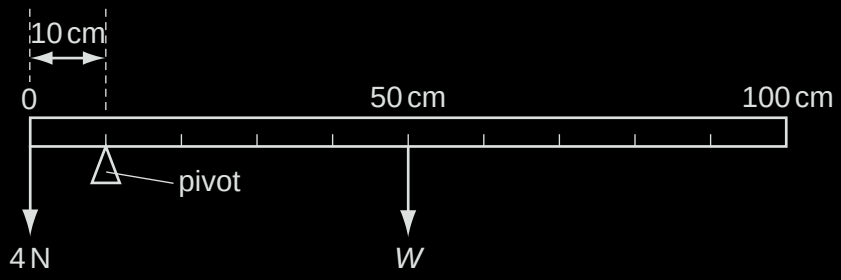
B the surface area of the reservoir

C the length of the reservoir

D the thickness of the dam wall

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What is the weight *W* of the metre rule?

A 1 N **B** 4 N **C** 16 N **D** 40 N

13 The tubes inside solar heating panels use the Sun's radiation to warm water.

Why are the tubes painted black?

A Black surfaces absorb radiation well.

B Black surfaces conduct heat well.

C Black surfaces emit radiation well.

D Black surfaces reflect radiation well.

14 A liquid evaporates rapidly.

Why does this cause it to cool?

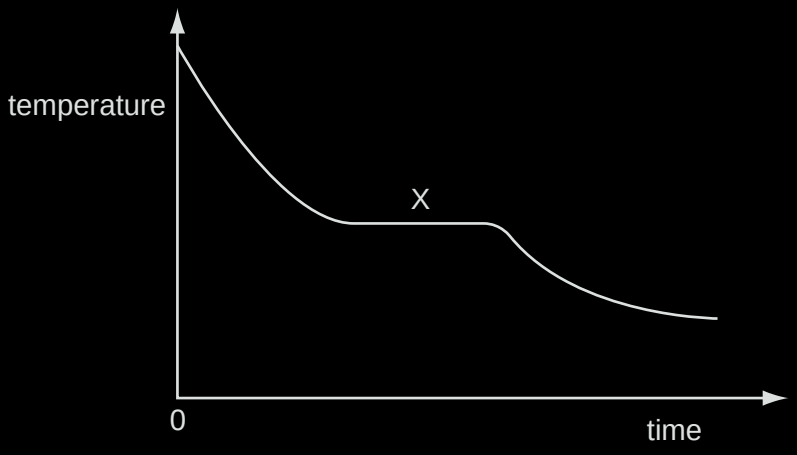
A Air molecules remove heat by contact with the liquid surface.

B Energy is lost by convection currents.

C Some of the most energetic molecules leave the liquid.

D The molecules have less room to move around.

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Which processes are taking place at region X?

A boiling and evaporation

B condensation only

C evaporation only

D solidification and evaporation

16 What is caused by the thermal expansion of a substance when heated?

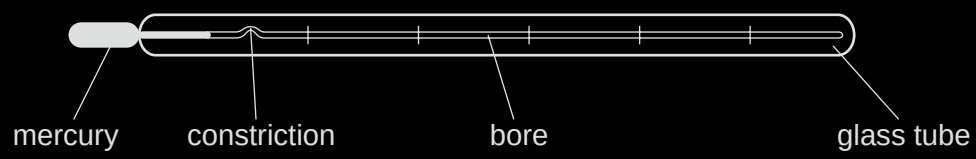
A a decrease in the resistance of a tungsten-filament lamp when switched on

B a rise in the pressure of the gas trapped in a gas cylinder placed in hot water

C the blowing of the fuse in a circuit when the current becomes too large

D the upward movement of the air above a Bunsen burner when it is lit

17 The diagram shows a clinical thermometer.



Which factor affects the sensitivity of the thermometer?

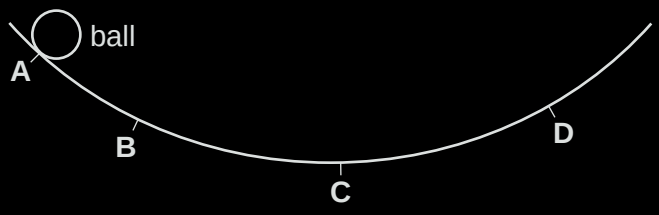
A the constriction

B the diameter of the bore

C the length of the glass tube

D the thickness of the glass tube

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21 During a thunderstorm, there is an interval of 1.70 s between an observer seeing the lightning and hearing the thunder. The speed of sound is 340 m / s.

What is the distance between the observer and the storm?

A 100 m **B** 200 m **C** 578 m **D** 1160 m

22 Which pair of emissions travels with the same speed in air?

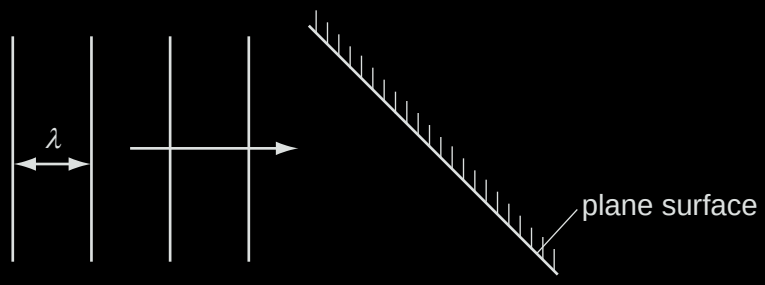
A alpha-particles and gamma-rays

B gamma-rays and infra-red waves

C infra-red waves and sound waves

D sound waves and alpha-particles

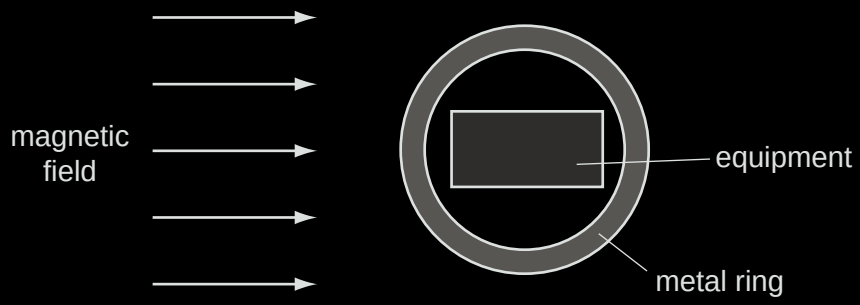
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Which row correctly describes the waves after they are reflected from the surface?

	speed of waves	wavelength λ
A	faster	shorter
B	slower	shorter
C	the same	longer
D	the same	the same

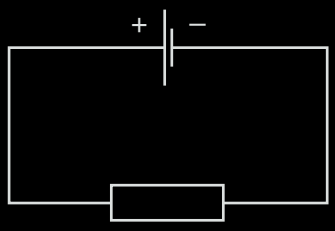
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Which metal should be used for the ring, and why?

	metal	reason
A	copper	the metal carries the field lines around the equipment
B	copper	the metal is non-magnetic
C	iron	the metal carries the field lines around the equipment
D	iron	the metal is non-magnetic

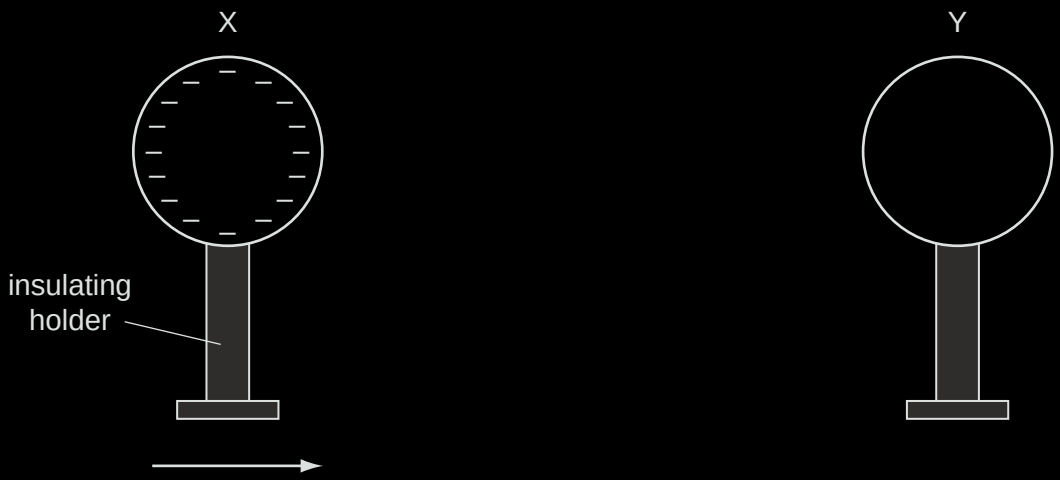
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Which row describes the charge on an electron and the direction of electron flow through the resistor?

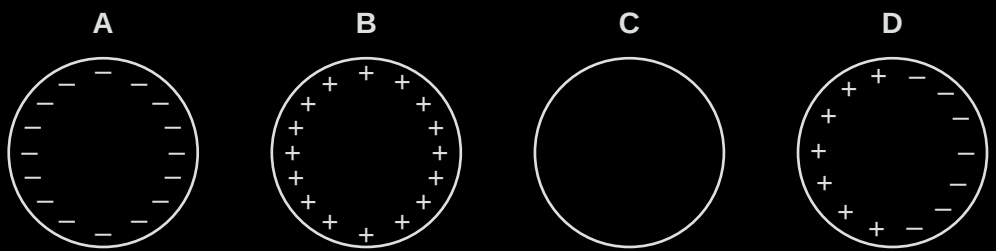
	charge on an electron	direction of electron flow
A	negative	– to +
B	negative	+ to –
C	positive	– to +
D	positive	+ to –

29 A negatively-charged sphere X is brought up to an identical uncharged sphere Y. The spheres do not touch.

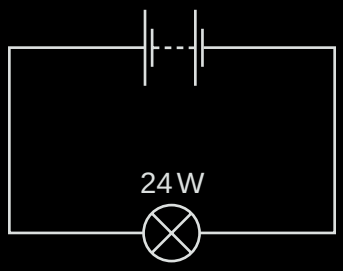


Sphere Y is 'earthed' by touching it with a finger, which is then removed. Sphere X is then moved away from sphere Y.

What is the final charge, if any, on sphere Y?



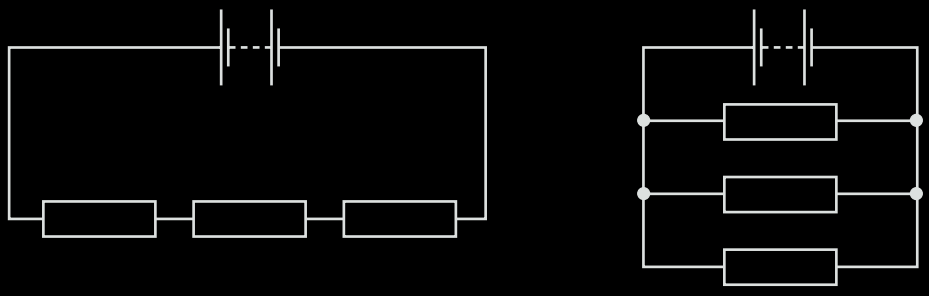
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What is the potential difference across the lamp?

A 5 V **B** 12 V **C** 24 V **D** 120 V

31 When three identical resistors are connected in series, their combined resistance is 6 Ω.

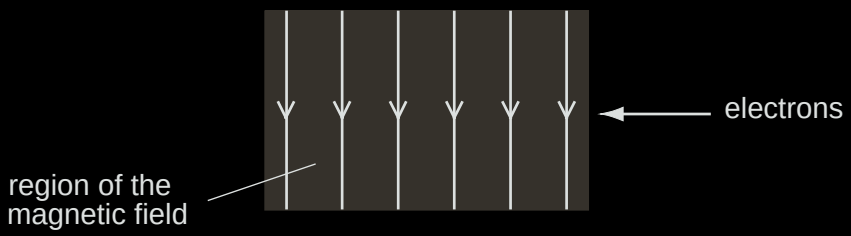


series parallel

What is their combined resistance when they are connected in parallel?

A $\frac{1}{6}$ Ω **B** $\frac{2}{3}$ Ω **C** $\frac{3}{2}$ Ω **D** 6 Ω

32 The diagram shows a beam of electrons entering a magnetic field. The direction of the magnetic field is downwards, towards the bottom of the page.



In which direction does the deflection of the electrons occur?

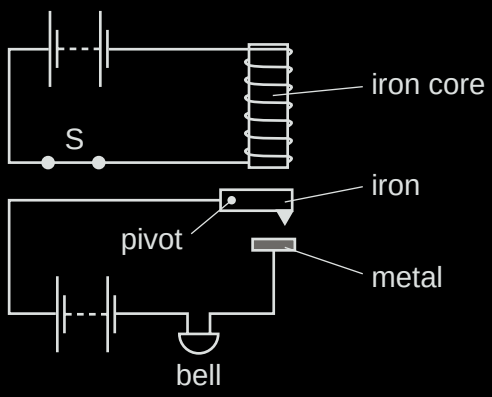
A into the page

B out of the page

C towards the bottom of the page

D towards the top of the page

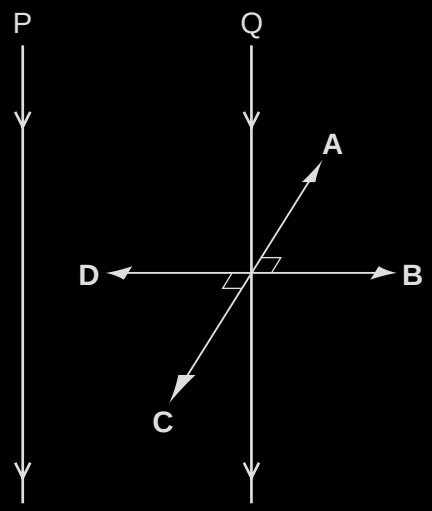
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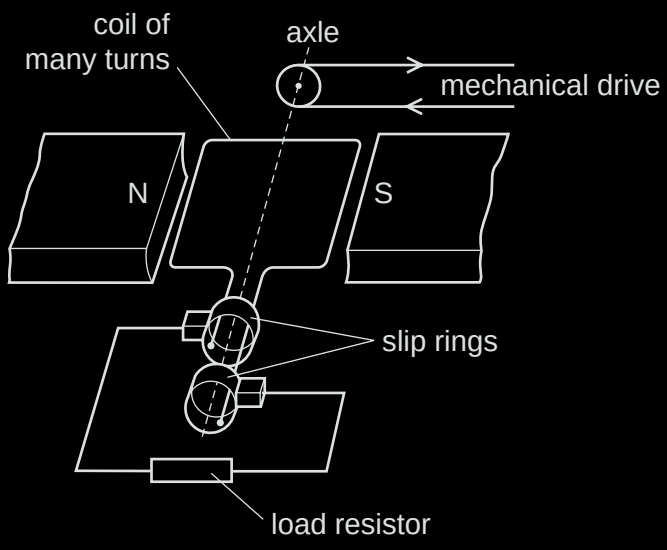
What happens when the switch S is opened?

	iron	bell
A	drops	rings
B	drops	stops ringing
C	moves up	rings
D	moves up	stops ringing

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37 The diagram shows an a.c. generator connected to an electrical circuit (load resistor).



Which statement is correct?

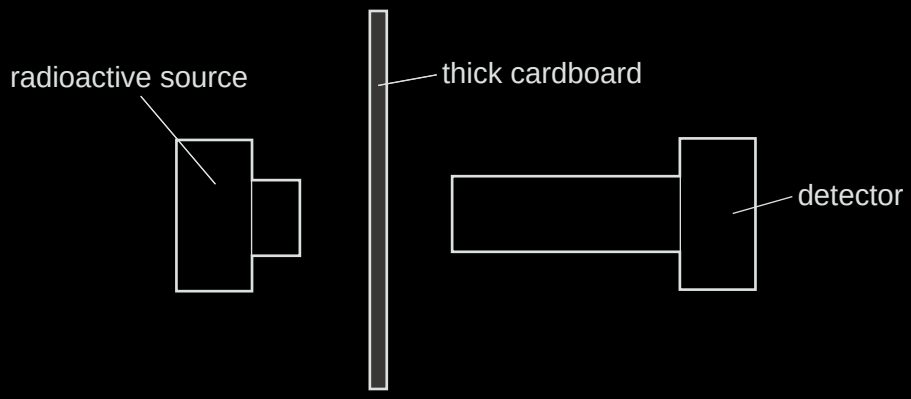
A The direction of the potential difference across the load resistor is always the same.

B The size of the induced e.m.f. depends on the number of turns in the coil.

C The size of the induced e.m.f. does not change as the coil turns.

D Winding the coil on a soft-iron cylinder makes no difference to the induced e.m.f.

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What is the reading on the detector now, and why?

	detector reading	reason
A	low	background radiation is detected
B	low	some alpha-particles pass through cardboard
C	zero	alpha-particles are all absorbed by the cardboard
D	zero	background radiation is all absorbed by the cardboard

40 $^{15}_{7}N$ is a nuclide of nitrogen.

How many electrons are there in a neutral atom of $^{15}_{7}N$?

A 7 **B** 8 **C** 15 **D** 22

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